

# CODE SPORT



In this month's column, we discuss a new natural language processing task for detecting adverse drug reactions from clinical/medical texts.

**N**atural language processing (NLP) has been applied in many different business domains such as law, transportation, customer care and healthcare. In last month's column, we discussed the problem of fact-checking, which is important in the political domain, wherein different politicians' claims are fact-checked for their truthfulness by automated techniques. In this month's column, we will look at the application of natural language processing to an important problem in the medical domain. This is the task of detecting adverse drug reactions automatically from text, with the use of NLP.

Medicines often have a number of side effects, quite apart from curing the disease they are supposed to address. Typically, these side effects are mild. Pharmaceutical companies are required by law to report any adverse reactions that could be encountered when a patient takes a drug manufactured by the firm. Known side effects experienced by patients during the clinical trials of the drug, along with their severity, are required to be recorded by pharma companies on the product's label.

Since different people react differently to the components of a drug, adverse reactions or side effects that are not known at the time of releasing the drug in the market, can still happen. Pharmacovigilance is the practice of monitoring the effects of medical drugs after they have been licensed for use, especially to identify and evaluate previously unreported adverse reactions. Given that some of the side effects can be harmful and, in certain cases, can even be life-threatening, pharma companies are required by law to provide post-marketing surveillance. This is the practice of monitoring the safety of a pharmaceutical drug or medical device after it has been released into the market and is an important part of the science of pharmacovigilance. There is a subtle difference between the two phrases, 'adverse drug reaction' and 'adverse drug events'.

An adverse drug reaction (ADR) is an injury caused by taking a medication. An adverse drug event (ADE) refers to any injury occurring at the time a drug is used, whether or not it is identified as a cause of the injury. Mentions of adverse drug reactions are typically found in clinical texts such as hospital clinical notes, discharge summaries, and physicians' notes. They can also be found in medical literature where clinical trials or studies of the drugs can contain such mentions. Of late, social media has been seen as another promising source of reporting adverse drug reactions. All these sources of ADR are text documents. Instead of manually analysing these documents for ADRs, one smart approach is to use text analytics to identify the presence of ADRs in these documents.

First, let us clearly define the task of detecting ADRs from text documents. We are given a set of documents that may contain ADRs. Our task is to identify all those present in the document. Each ADR can be represented by the tuple *<Drug name, side effect>*. Since an adverse reaction can occur sometimes only due to overdosing, it is possible to expand the above representation of ADR to contain the dosage as an optional component. For now, let us confine ourselves to detecting the drug and the side effect caused by it as the components of ADR.

A clever reader may now be wondering whether unintended side effects caused by drugs are always negative, and whether there could also be positive side effects. Yes, those too, do occur and it would be good to capture them also under the ADR umbrella.

Let us consider a few examples of ADRs.

*E1: "Prozac has definitely helped reduce my anxiety, but I have been having severe stomach cramps."*

In this example, the text contains an ADR *<Prozac, stomach cramp>*. But you may be wondering whether *<Prozac, anxiety>* should also be marked as an